

 STOCKWHISK SAAS PLATFORM

DOC-16

Database Schema & Entity-Relationship (ER) Specification

Complete 54-Model Multi-Tenant Relational Schema, Mermaid ERDs & Data Dictionary

DOCUMENT TYPE	TARGET AUDIENCE	RELEASE VERSION	STATUS
Database Architecture	Database Architects, Backend Engineers, DevOps	v1.4.0 Production	✅ Verified & Approved

16. StockWhisk Database Schema & Entity-Relationship (ER) Specification

1. Executive Summary & Architecture Overview

The StockWhisk relational database architecture is engineered as a **Multi-Tenant SaaS Retail ERP, POS, RMA Service, and Financial Accounting engine**. It is built on Django ORM with native support for PostgreSQL (Production), MySQL, and SQLite (Local Development).

Core Architectural Principles:

1. **Tenant Isolation by Construction:** All business entities inherit from `TenantScopedModel` (via `core.models.TenantScopedModel`), which automatically injects a `shop_id` foreign key and enforces tenant query filtering through `TenantManager`.
2. **Immutable Financial & Stock Ledgers (Event-Sourced):**
 - Cash/Bank movements are logged into the append-only `LedgerEntry` table.
 - Physical warehouse movements are recorded in the append-only `StockMovement` table.
 - Derived quantities (`Product.current_stock`, `Customer.due_balance`, `Supplier.due_balance`) are cached projections protected by database constraints and atomic transactions.
3. **Dual-Layer Inventory Tracking:**
 - Standard bulk inventory tracked by quantity.
 - **HIGH**-value serialized inventory tracked per individual physical piece (`ProductUnit`) with unique barcodes and serialized lifecycle states.

2. HIGH -Level Entity-Relationship (ER) Diagrams

2.1 Core Multi-Tenancy, Auth & Subscriptions ERD

```
erDiagram
    SubscriptionPlan ||--o{ Shop : "subscribed_under"
    SubscriptionPlan ||--o{ Subscription : "defines_terms"
    Shop ||--o{ Subscription : "has_history"
    Shop ||--o{ Branch : "operates"
    Shop ||--o{ Role : "defines_roles"
    Shop ||--o{ User : "employs"
    Shop ||--o{ APIKey : "provisions"
    Branch ||--o{ User : "assigned_to"
    Role }|--}|{ Permission : "grants"
    User ||--o{ Notification : "receives"
    Subscription ||--o{ SubscriptionInvoice : "bills"
    SubscriptionInvoice ||--o{ ManualPayment : "settled_by"

    Shop {
        int id PK
        string name
        string slug
        string subdomain
        string currency
        decimal vat_percent
        boolean emi_enabled
        boolean is_active
    }

    Branch {
        int id PK
        int shop_id FK
        string name
        string code
        boolean is_main
    }

    User {
        int id PK
        int shop_id FK
        int branch_id FK
        string email
        string role
        string phone
        datetime last_seen
    }

    SubscriptionPlan {
        int id PK
        string name
        string tier
        decimal price_monthly
        decimal price_yearly
        int max_users
        int max_branches
        int max_products
    }
```

2.2 Product Catalog, Purchasing & Stock Movements ERD

```
erDiagram
    Shop ||--o{ Category : "categorizes"
    Shop ||--o{ Brand : "brands"
    Shop ||--o{ Product : "owns"
    Category ||--o{ Product : "classifies"
    Brand ||--o{ Product : "brands"
    Supplier ||--o{ Product : "primary_supplier"
    Product ||--o{ ProductVariation : "has_variants"
    Product ||--o{ ProductUnit : "tracks_serials"
    Product ||--o{ StockMovement : "logs_movements"
    Branch ||--o{ StockMovement : "movement_location"
    Supplier ||--o{ PurchaseOrder : "receives_from"
    PurchaseOrder ||--o{ PurchaseOrderItem : "contains"
    Product ||--o{ PurchaseOrderItem : "ordered_item"
    PurchaseOrder ||--o{ PurchasePayment : "paid_via"
    Supplier ||--o{ SupplierPayment : "settles_due"

    Product {
        int id PK
        int shop_id FK
        string name
        string sku
        string barcode
        decimal cost_price
        decimal selling_price
        decimal current_stock
        boolean track_inventory
        int reorder_level
    }

    ProductUnit {
        int id PK
        int product_id FK
        string barcode UK
        string status
        decimal cost_price
        decimal selling_price
        int warranty_months
        int sale_id FK
    }

    StockMovement {
        int id PK
        int product_id FK
        int branch_id FK
        string movement_type
        decimal quantity
        decimal unit_cost
        string reference_type
        string reference_id
    }
```

2.3 POS Sales, Invoicing, Returns, CRM & EMI ERD

```

erDiagram
    Shop ||--o{ Customer : "manages"
    Customer ||--o{ Sale : "purchases"
    Customer ||--o{ CustomerPayment : "pays_due"
    Customer ||--o{ EMISchedule : "finances"
    Branch ||--o{ Sale : "originates"
    Sale ||--o{ SaleItem : "contains"
    Product ||--o{ SaleItem : "sold_product"
    Sale ||--o{ Payment : "settled_with"
    Sale ||--o{ SaleReturn : "returned_in"
    SaleReturn ||--o{ SaleReturnItem : "contains_items"
    Sale ||--o{ EMISchedule : "installment_plan"
    EMISchedule ||--o{ EMIInstallment : "monthly_dues"

    Sale {
        int id PK
        int shop_id FK
        int customer_id FK
        string invoice_no UK
        datetime sale_date
        decimal subtotal
        decimal discount
        decimal delivery_charge
        decimal tax
        decimal total
        decimal paid
        string status
    }

    SaleItem {
        int id PK
        int sale_id FK
        int product_id FK
        decimal quantity
        decimal unit_price
        decimal unit_cost
        decimal discount
        decimal subtotal
    }

    EMISchedule {
        int id PK
        int sale_id FK
        int customer_id FK
        decimal total_emi_amount
        decimal down_payment
        decimal interest_percent
        int total_months
        decimal monthly_installment
        string status
    }

    EMIInstallment {
        int id PK
        int schedule_id FK
        int installment_number
        date due_date
        decimal amount
        decimal paid_amount
  
```

```
    string status  
  }
```

2.4 Financial Accounting, Daily Settlement & Cash Flow ERD

```
erDiagram
    Shop ||--o{ ExpenseCategory : "categorizes"
    ExpenseCategory ||--o{ Expense : "classifies"
    Shop ||--o{ Expense : "incurs"
    Shop ||--o{ LedgerEntry : "records_cashflow"
    Shop ||--o{ DailySettlement : "reconciles"
    Shop ||--o{ Investment : "equity_movements"
    Shop ||--o{ AccountTransfer : "liquid_transfers"

    LedgerEntry {
        int id PK
        int shop_id FK
        string account
        decimal amount
        string source_type
        string source_id
        string description
        datetime created_at
    }

    DailySettlement {
        int id PK
        int shop_id FK
        datetime opened_at
        datetime closed_at
        decimal opening_cash
        decimal expected_cash
        decimal actual_cash
        decimal discrepancy
        decimal total_sales
        decimal total_expenses
        decimal total_refunds
        string status
    }

    Investment {
        int id PK
        int shop_id FK
        string investor_name
        string type
        decimal amount
        date invested_on
        string payment_method
    }

    AccountTransfer {
        int id PK
        int shop_id FK
        string from_account
        string to_account
        decimal amount
        date transferred_on
        string reference
    }
```

2.5 RMA Warranty & Repair Services ERD

```

erDiagram
    Product ||--o{ Warranty : "covered_by"
    ProductUnit ||--o{ Warranty : "bound_to"
    Customer ||--o{ Warranty : "registered_to"
    Warranty ||--o{ WarrantyClaim : "claims"
    Warranty ||--o{ ServiceTicket : "service_job"
    Customer ||--o{ ServiceTicket : "customer_device"
    User ||--o{ ServiceTicket : "assigned_technician"
    ServiceTicket ||--o{ ServiceTicketPart : "parts_consumed"
    Product ||--o{ ServiceTicketPart : "inventory_part"
    ServiceTicket ||--o{ ServiceTicketStatusHistory : "audit_log"

    Warranty {
        int id PK
        int product_id FK
        int product_unit_id FK
        int customer_id FK
        string serial_no
        int period_months
        date start_date
        date expiry_date
        string status
    }

    ServiceTicket {
        int id PK
        int shop_id FK
        int customer_id FK
        int warranty_id FK
        int technician_id FK
        string ticket_no UK
        string device_type
        string issue_type
        string status
        decimal service_charge
        decimal discount
        decimal paid
    }

    ServiceTicketPart {
        int id PK
        int ticket_id FK
        int product_id FK
        decimal quantity
        decimal unit_cost
        decimal unit_price
        boolean from_stock
    }

```

3. Complete Data Dictionary (Table by Table)

3.1 App: tenants

Table: tenants_subscriptionplan

Defines global SaaS billing tiers, feature quotas, and limits.

COLUMN	TYPE	NULLABLE	CONSTRAINTS	DESCRIPTION
id	BigAutoField	No	PK	Primary Key
name	CharField(100)	No		Display name (e.g. Free, Basic, Pro, Enterprise)
tier	CharField(20)	No	Enum (FREE , BASIC , PROFESSIONAL , ENTERPRISE)	Plan tier identifier
price_monthly	DecimalField(10,2)	No	Default 0.00	Monthly billing price
price_yearly	DecimalField(10,2)	No	Default 0.00	Yearly billing price
features	JSONField	No	Default list	Active feature toggles
max_users	PositiveIntegerField	No	Default 1	Max user limit
max_branches	PositiveIntegerField	No	Default 1	Max branches allowed
max_products	PositiveIntegerField	No	Default 100	Max product catalog size
is_active	BooleanField	No	Default True	Whether plan is purchasable

Table: tenants_shop

The root tenant partition representing an independent business.

COLUMN	TYPE	NULLABLE	CONSTRAINTS	DESCRIPTION
id	BigAutoField	No	PK	Primary Key
name	CharField(150)	No		Legal / Commercial business name
slug	SlugField(160)	No	Unique	URL slug
subdomain	CharField(63)	Yes	Unique	Optional dedicated subdomain
logo	ImageField	Yes		Uploaded business brand logo
phone	CharField(20)	No		Primary contact phone
email	EmailField	Yes		Primary business notification email
currency	CharField(10)	No	Default 'BDT'	Currency code (BDT, USD, etc.)
vat_enabled	BooleanField	No	Default False	Whether VAT is charged on sales
vat_percent	DecimalField(5,2)	No	Default 0.00	Percentage rate of VAT
emi_enabled	BooleanField	No	Default True	Enable installment sales
offline_sale_mode	BooleanField	No	Default False	Enable offline cached POS sync
plan_id	ForeignKey	Yes	FK -> tenants_subscriptionplan	Current active subscription plan
trial_ends_at	DateTimeField	Yes		Free trial expiration timestamp
is_active	BooleanField	No	Default True	Active shop status
is_demo	BooleanField	No	Default False	Public read-only demo shop
is_free	BooleanField	No	Default False	Lifetime free granted shop

Table: tenants_branch

Physical outlet / warehouse branches under a single tenant shop.

COLUMN	TYPE	NULLABLE	CONSTRAINTS	DESCRIPTION
id	BigAutoField	No	PK	Primary Key
shop_id	ForeignKey	No	FK -> tenants_shop	Parent tenant shop
name	CharField(100)	No		Branch name (e.g. Main Branch, Gulshan Outlet)
code	CharField(20)	Yes		Unique branch short code
phone	CharField(20)	Yes		Branch telephone / mobile
address	TextField	Yes		Physical location address
is_main	BooleanField	No	Default False	Designates primary headquarters branch
is_active	BooleanField	No	Default True	Branch operational status

3.2 App: accounts

Table: accounts_user

System user accounts (Cashiers, Managers, Owners, Resellers, Platform Superadmins).

COLUMN	TYPE	NULLABLE	CONSTRAINTS	DESCRIPTION
id	BigAutoField	No	PK	Primary Key
email	EmailField	No	Unique	Unique login identifier
shop_id	ForeignKey	Yes	FK -> tenants_shop	Scoped tenant shop (null for platform admin)
branch_id	ForeignKey	Yes	FK -> tenants_branch	Scoped branch (null for all-branch managers)
role	CharField(20)	No	Enum (OWNER , MANAGER , CASHIER , STAFF , RESELLER , PLATFORM_ADMIN)	User role type
phone	CharField(20)	Yes		Contact mobile number
is_staff	BooleanField	No	Default False	Platform administration access
last_seen	DateTimeField	Yes		Timestamp of last user API activity

Table: accounts_role & accounts_permission

Granular role-based access control (RBAC).

TABLE	KEY COLUMNS	DESCRIPTION
accounts_permission	code (Unique), name , category , description	Global catalog of permissions (e.g. create_sale , manage_expenses)
accounts_role	shop_id , name , role_type , is_system	Tenant-defined customizable permission roles
accounts_role_permissions	role_id (FK), permission_id (FK)	Many-to-Many junction table for RBAC

3.3 App: catalog

Table: catalog_product

Master physical and service product inventory items.

COLUMN	TYPE	NULLABLE	CONSTRAINTS	DESCRIPTION
id	BigAutoField	No	PK	Primary Key
shop_id	ForeignKey	No	FK -> tenants_shop	Tenant owner
name	CharField(200)	No		Product title / model name
sku	CharField(100)	Yes	Index	Stock Keeping Unit
barcode	CharField(100)	Yes	Index	Primary retail barcode (EAN-13, Code128, etc.)
category_id	ForeignKey	Yes	FK -> catalog_category	Product category classification
brand_id	ForeignKey	Yes	FK -> catalog_brand	Brand / Manufacturer
unit_id	ForeignKey	Yes	FK -> catalog_unit	Unit of measure (Pcs, Box, Kg, Liter)
supplier_id	ForeignKey	Yes	FK -> purchasing_supplier	Default vendor
cost_price	DecimalField(12,2)	No	Default 0.00	Latest unit purchase cost price
selling_price	DecimalField(12,2)	No	Default 0.00	Standard retail selling price
track_inventory	BooleanField	No	Default True	If False, item is untracked service/fee
current_stock	DecimalField(12,2)	No	Default 0.00	Cached on-hand inventory quantity
reorder_level	PositiveIntegerField	No	Default 5	Safety stock threshold for alerts
warranty_months	PositiveSmallInt	No	Default 0	Standard warranty duration in months

COLUMN	TYPE	NULLABLE	CONSTRAINTS	DESCRIPTION
replacement_guarantee_days	PositiveSmallInt	No	Default 0	Replacement coverage in days

Table: catalog_productunit

HIGH -precision serialized unit tracking (IMEI, unique serial numbers).

COLUMN	TYPE	NULLABLE	CONSTRAINTS	DESCRIPTION
id	BigAutoField	No	PK	Primary Key
shop_id	ForeignKey	No	FK -> tenants_shop	Tenant owner
product_id	ForeignKey	No	FK -> catalog_product	Parent catalog product
barcode	CharField(100)	No	Unique per product	Unique serial / IMEI barcode
status	CharField(20)	No	Enum (IN_STOCK , SOLD , TESTING_PENDING , DEFECTIVE , RETURNED_SUPPLIER)	Physical unit status
cost_price	DecimalField(12,2)	No		Snapshoted purchase cost of this unit
selling_price	DecimalField(12,2)	No		Snapshoted selling price
sale_id	ForeignKey	Yes	FK -> sales_sale	Linked sale invoice when sold
sold_at	DateTimeField	Yes		Timestamp when unit was sold

3.4 App: inventory

Table: inventory_stockmovement

Immutable, append-only warehouse stock ledger.

COLUMN	TYPE	NULLABLE	CONSTRAINTS	DESCRIPTION
id	BigAutoField	No	PK	Primary Key
shop_id	ForeignKey	No	FK -> tenants_shop	Tenant owner
product_id	ForeignKey	No	FK -> catalog_product	Modified product
branch_id	ForeignKey	Yes	FK -> tenants_branch	Branch location where stock changed
movement_type	CharField(30)	No	Enum (PURCHASE_IN , SALE_OUT , SALE_RETURN_IN , ADJUSTMENT_IN , ADJUSTMENT_OUT , DAMAGE_OUT , TRANSFER_IN , TRANSFER_OUT , SERVICE_CONSUMED)	Nature of stock change
quantity	DecimalField(12,2)	No		Positive or negative delta quantity
unit_cost	DecimalField(12,2)	No		Snapshoted unit cost at movement time
reference_type	CharField(50)	Yes		Source model name (e.g. 'Sale' , 'PurchaseOrder')
reference_id	CharField(100)	Yes		Source transaction ID or invoice number
created_by_id	ForeignKey	Yes	FK -> accounts_user	Operator who executed the movement
created_at	DateTimeField	No	Auto Now Add	Movement execution timestamp

3.5 App: sales & pos

Table: sales_sale

Master sales invoice document.

COLUMN	TYPE	NULLABLE	CONSTRAINTS	DESCRIPTION
id	BigAutoField	No	PK	Primary Key
shop_id	ForeignKey	No	FK -> tenants_shop	Tenant owner
invoice_no	CharField(50)	No	Unique per shop	System generated memo number
customer_id	ForeignKey	Yes	FK -> crm_customer	Registered customer (or null for walk-in)
customer_name	CharField(150)	Yes		Walk-in buyer name
customer_phone	CharField(20)	Yes		Walk-in buyer contact number
sale_date	DateTimeField	No	Index	Timestamp of checkout
subtotal	DecimalField(12,2)	No	Default 0.00	Gross sum of all items
discount	DecimalField(12,2)	No	Default 0.00	Total discount deducted
delivery_charge	DecimalField(12,2)	No	Default 0.00	Shipping / courier fee
tax	DecimalField(12,2)	No	Default 0.00	Computed VAT amount
total	DecimalField(12,2)	No	Default 0.00	Net payable: subtotal - discount + delivery + tax
paid	DecimalField(12,2)	No	Default 0.00	Total cash/digital amount received
status	CharField(20)	No	Enum (PAID , PARTIAL , DUE , CANCELLED , QUOTATION)	Invoicing payment status
idempotency_key	CharField(100)	Yes	Unique	Prevents duplicate POS checkouts

Table: sales_saleitem

Individual product line items within a sales invoice.

COLUMN	TYPE	NULLABLE	CONSTRAINTS	DESCRIPTION
id	BigAutoField	No	PK	Primary Key
sale_id	ForeignKey	No	FK -> sales_sale	Parent sales invoice
product_id	ForeignKey	No	FK -> catalog_product	Sold catalog product
quantity	DecimalField(12,2)	No		Quantity sold
unit_price	DecimalField(12,2)	No		Selling price charged per unit
unit_cost	DecimalField(12,2)	No		Snapshoted purchase cost (for COGS & profit)
discount	DecimalField(12,2)	No	Default 0.00	Item level discount
subtotal	DecimalField(12,2)	No		Net line amount

Table: sales_emischedule & sales_emiinstallment

Financed installment sales plans.

COLUMN	TYPE	NULLABLE	CONSTRAINTS	DESCRIPTION
id	BigAutoField	No	PK	Primary Key
sale_id	OneToOneField	No	FK -> sales_sale	Linked sale invoice
customer_id	ForeignKey	No	FK -> crm_customer	Financed customer
total_emi_amount	DecimalField(12,2)	No		Principal + total interest
down_payment	DecimalField(12,2)	No		Upfront initial cash paid
interest_percent	DecimalField(5,2)	No	Default 0.00	Interest rate charged
total_months	PositiveSmallInt	No		Number of installment months
monthly_installment	DecimalField(12,2)	No		Monthly amount due
status	CharField(20)	No	Enum (ACTIVE , COMPLETED , DEFAULTED)	EMI schedule health

3.6 App: accounting

Table: accounting_ledgerentry

Append-only double-entry financial cash flow tracking table.

COLUMN	TYPE	NULLABLE	CONSTRAINTS	DESCRIPTION
id	BigAutoField	No	PK	Primary Key
shop_id	ForeignKey	No	FK -> tenants_shop	Tenant owner
account	CharField(20)	No	Enum (CASH , BKASH , NAGAD , BANK , CARD , OTHER)	Specific liquid account
amount	DecimalField(12,2)	No		Positive (Inflow) or Negative (Outflow)
source_type	CharField(50)	No		Event type (SALE , EXPENSE , DUE_COLLECT , PURCHASE , TRANSFER , INVESTMENT , DRAWING)
source_id	CharField(100)	Yes		Foreign source record ID
description	CharField(255)	Yes		Human-readable audit narrative
created_at	DateTimeField	No	Auto Now Add	Financial entry timestamp

Table: accounting_dailysettlement

Daily shift reconciliation and cash drawer closing audits.

COLUMN	TYPE	NULLABLE	CONSTRAINTS	DESCRIPTION
id	BigAutoField	No	PK	Primary Key
shop_id	ForeignKey	No	FK -> tenants_shop	Tenant owner
opened_at	DateTimeField	No		Shift opening timestamp
closed_at	DateTimeField	Yes		Shift closing timestamp
opening_cash	DecimalField(12,2)	No		Carryover cash from previous shift
expected_cash	DecimalField(12,2)	No		System calculated cash: opening + sales - expenses
actual_cash	DecimalField(12,2)	Yes		Manually counted cash in physical drawer
discrepancy	DecimalField(12,2)	Yes		Variance: actual_cash - expected_cash
status	CharField(20)	No	Enum (OPEN , CLOSED)	Shift register state

3.7 App: service

Table: service_warranty & service_serviceticket

Warranty lifecycle and repair center job card management.

TABLE	KEY COLUMNS	DESCRIPTION
service_warranty	product_id , product_unit_id , customer_id , serial_no , period_months , start_date , expiry_date , status	Warranty registration bound to individual serial units.
service_serviceticket	ticket_no , customer_id , warranty_id , technician_id , device_type , issue_type , complaint , status , service_charge , discount , paid	Service job card state machine (RECEIVED → DIAGNOSING → AWAITING_PARTS → IN_REPAIR → READY → DELIVERED).
service_serviceticketpart	ticket_id , product_id , quantity , unit_cost , unit_price , from_stock	Replacement spare parts used in repair with optional automated inventory deduction.

4. Key Database Constraints & Integrity Rules

```
-- 1. Strict Tenant Scoping Unique Constraints
ALTER TABLE catalog_product
ADD CONSTRAINT uniq_sku_per_shop_when_set
UNIQUE (shop_id, sku);

ALTER TABLE catalog_productunit
ADD CONSTRAINT uniq_unit_barcode_per_product
UNIQUE (shop_id, product_id, barcode);

ALTER TABLE sales_sale
ADD CONSTRAINT uniq_invoice_per_shop
UNIQUE (shop_id, invoice_no);

ALTER TABLE service_serviceticket
ADD CONSTRAINT uniq_ticket_no_per_shop
UNIQUE (shop_id, ticket_no);

-- 2. Non-Negative Financial Amounts (Check Constraints)
ALTER TABLE sales_payment
ADD CONSTRAINT payment_amount_positive
CHECK (amount > 0);

ALTER TABLE accounting_expense
ADD CONSTRAINT expense_amount_positive
CHECK (amount ≥ 0);

ALTER TABLE accounting_investment
ADD CONSTRAINT investment_amount_positive
CHECK (amount ≥ 0);
```

5. Summary Index Statistics

APP AREA	TOTAL MODELS	CORE DATABASE TABLES	KEY FOREIGN KEYS	UNIQUE CONSTRAINTS
Tenants & Core	4	4	6	3
Auth & Accounts	5	6	4	2
Catalog & Products	6	6	8	5
Inventory & Movements	1	1	3	1
CRM & Customers	2	2	2	1
Purchasing & Vendors	5	5	7	2
Sales, POS & EMI	7	7	10	4
Finance & Accounting	7	7	5	3
Service & Warranty	5	5	8	2
Platform & Billing	12	12	6	2
TOTALS	54 Models	55 Tables	59 Relations	25 Constraints